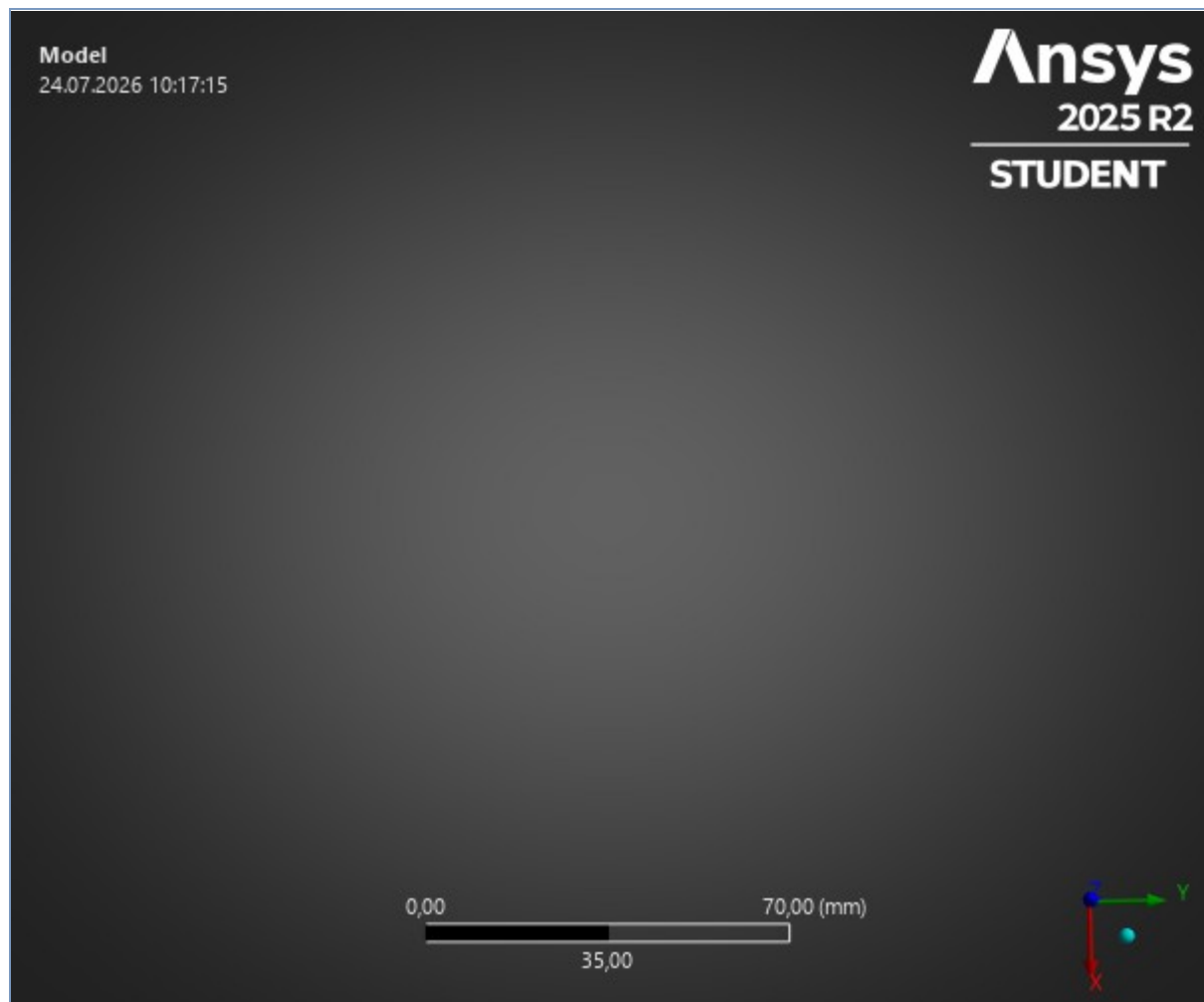




Project*

First Saved	Monday, July 20, 2026
Last Saved	Thursday, July 23, 2026
Product Version	2025 R2
Save Project Before Solution	No
Save Project After Solution	No



Contents

- [Units](#)
- [Model \(B4\)](#)
 - [Geometry Imports](#)
 - [Geometry Import \(B3\)](#)
 - [Geometry](#)
 - [max_pressure|Вырез-Вытянуть2](#)
 - [Materials](#)
 - [Coordinate Systems](#)
 - [Connections](#)
 - [Mesh](#)
 - [Face Sizing](#)
 - [Named Selections](#)
 - [Static Structural \(B5\)](#)
 - [Analysis Settings](#)
 - [Rotational Velocity](#)
 - [Loads](#)
 - [Solution \(B6\)](#)
 - [Solution Information](#)
 - [Results](#)
 - [Stress Tool](#)
 - [Safety Factor](#)
- [Material Data](#)
 - [PA12-CF10](#)

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (B4)

TABLE 2

Model (B4) > Geometry Imports

Object Name	<i>Geometry Imports</i>
State	Solved

TABLE 3

Model (B4) > Geometry Imports > Geometry Import (B3)

Object Name	<i>Geometry Import (B3)</i>
State	Solved
Definition	
Source	C:\Users\User\Desktop\Прочностные характеристики\max_pressure\max_pressure.step
Type	Step
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes

Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (B4) > Geometry

Object Name	<i>Geometry</i>
State	Fully Defined
Definition	
Source	C:\Users\User\Desktop\Прочностные характеристики\max_pressure\max_pressure.step
Type	Step
Length Unit	Millimeters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	200, mm
Length Y	200, mm
Length Z	51,025 mm
Properties	
Volume	6,9549e+005 mm ³
Mass	0,72331 kg
Scale Factor Value	1,
Statistics	
Bodies	1
Active Bodies	1
Nodes	127598
Elements	72770
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No

Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (B4) > Geometry > Parts

Object Name	<i>max_pressure Вырез-Вытянуть2</i>
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	PA12-CF10
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	200, mm
Length Y	200, mm
Length Z	51,025 mm
Properties	
Volume	6,9549e+005 mm ³
Mass	0,72331 kg
Centroid X	-4,5014e-003 mm
Centroid Y	9,8156e-004 mm
Centroid Z	13,043 mm
Moment of Inertia Ip1	1361,4 kg·mm ²
Moment of Inertia Ip2	1361, kg·mm ²
Moment of Inertia Ip3	2513,2 kg·mm ²
Statistics	
Nodes	127598
Elements	72770
Mesh Metric	None

TABLE 6
Model (B4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	4
Material Assignments	0

Coordinate Systems

TABLE 7
Model (B4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0,
Origin	
Origin X	0, mm
Origin Y	0, mm
Origin Z	0, mm
Directional Vectors	
X Axis Data	[1, 0, 0,]
Y Axis Data	[0, 1, 0,]
Z Axis Data	[0, 0, 1,]
Transfer Properties	
Source	
Read Only	No

Connections

TABLE 8
Model (B4) > Connections

Object Name	<i>Connections</i>
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	0
Active Contacts	0
Joints	0
Active Joints	0
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0
Body Interactions	0
Active Body Interactions	0

Mesh

TABLE 9

Model (B4) > Mesh	
Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	3,39 mm
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	287,41 mm
Average Surface Area	531,5 mm ²
Minimum Edge Length	7,3636e-004 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5,e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0,272
Maximum Layers	5
Growth Rate	1,2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	127598
Elements	72770
Show Detailed Statistics	No

TABLE 10
Model (B4) > Mesh > Mesh Controls

Object Name	<i>Face Sizing</i>
State	Fully Defined
Scope	

Scoping Method	Geometry Selection
Geometry	78 Faces
Definition	
Suppressed	No
Type	Element Size
Element Size	2,8 mm
Advanced	
Defeature Size	Default
Influence Volume	No
Behavior	Soft

Named Selections

Static Structural (B5)

TABLE 11
Model (B4) > Analysis

Object Name	<i>Static Structural (B5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22, °C
Generate Input Only	No

TABLE 12
Model (B4) > Static Structural (B5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	
Number Of Steps	1,
Current Step Number	1,
Step End Time	1, s
Auto Time Stepping	On
Define By	Substeps
Initial Substeps	20,
Minimum Substeps	10,
Maximum Substeps	100,
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Program Controlled
Solver Pivot Checking	Program Controlled
Large Deflection	On
Inertia Relief	Off
Quasi-Static Solution	On
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled

Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Constant
--Method	Energy
--Energy Dissipation Ratio	1,e-004
--Activation For First Substep	No
--Stabilization Force Limit	0,2
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\User\Desktop\Прочностные характеристики\All_work_pressure_files\dp0\SYS-1\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes
Solver Units	Active System
Solver Unit System	nmm

TABLE 13
Model (B4) > Static Structural (B5) > Rotations

Object Name	<i>Rotational Velocity</i>
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Define By	Components
Coordinate System	Global Coordinate System
X Component	0, rad/s (ramped)
Y Component	0, rad/s (ramped)
Z Component	-1256,6 rad/s (ramped)

X Coordinate	5,e-004 mm
Y Coordinate	-6,1961e-004 mm
Z Coordinate	4,65e-002 mm
Suppressed	No

FIGURE 1
Model (B4) > Static Structural (B5) > Rotational Velocity

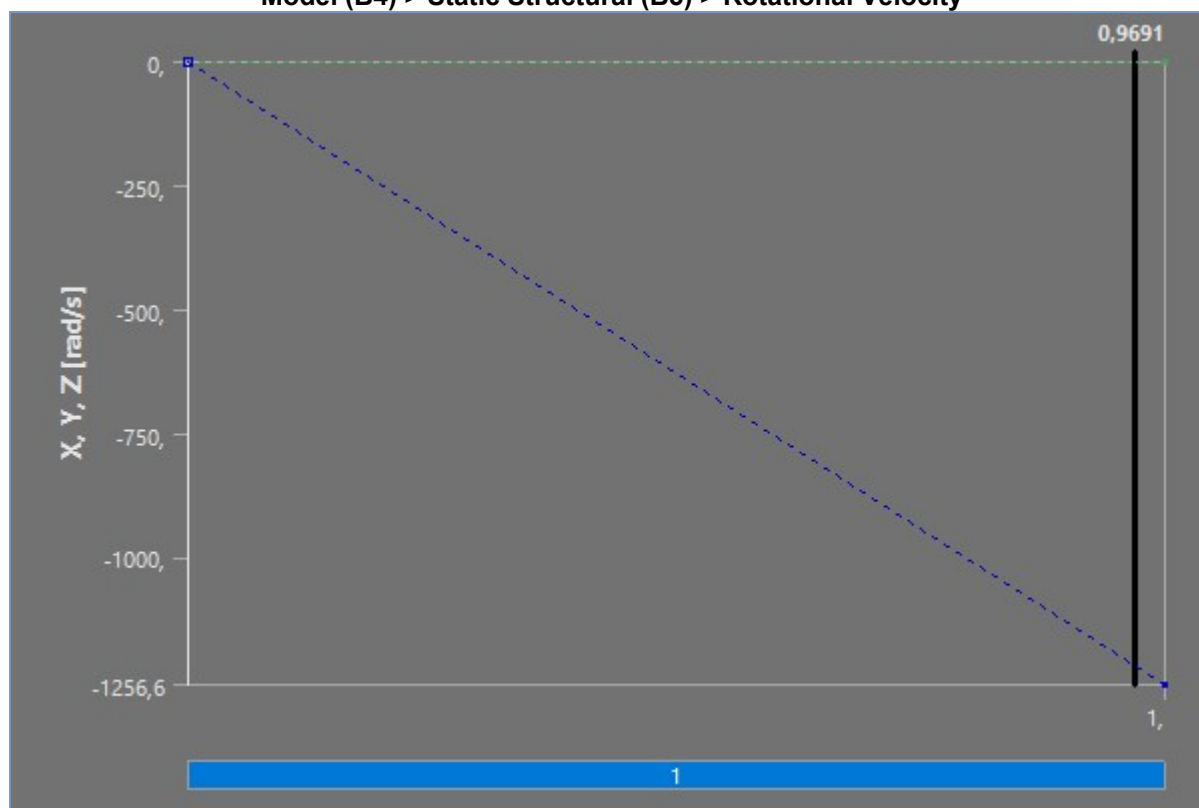


TABLE 14
Model (B4) > Static Structural (B5) > Loads

Object Name	Thermal Condition	Pressure	Cylindrical Support	Remote Displacement
State	Fully Defined			Suppressed
Scope				
Scoping Method	Geometry Selection			
Geometry	1 Body	78 Faces	2 Faces	
Coordinate System				Global Coordinate System
X Coordinate				2,7818e-016 mm
Y Coordinate				2,6427e-016 mm
Z Coordinate				21,75 mm
Location				Defined
Definition				
Type	Thermal Condition	Pressure	Cylindrical Support	Remote Displacement
Magnitude	75, °C (ramped)	0,10132 MPa (ramped)		
Suppressed	No			Yes
Define By		Normal To		
Applied By		Surface Effect		
Loaded Area		Deformed		
Radial			Fixed	
Axial			Fixed	
Tangential			Free	

X Component		0, mm (ramped)
Y Component		0, mm (ramped)
Z Component		0, mm (ramped)
Rotation X		0, ° (ramped)
Rotation Y		0, ° (ramped)
Rotation Z		0, ° (ramped)
Behavior		Deformable
Advanced		
Pinball Region		All

FIGURE 2
Model (B4) > Static Structural (B5) > Thermal Condition

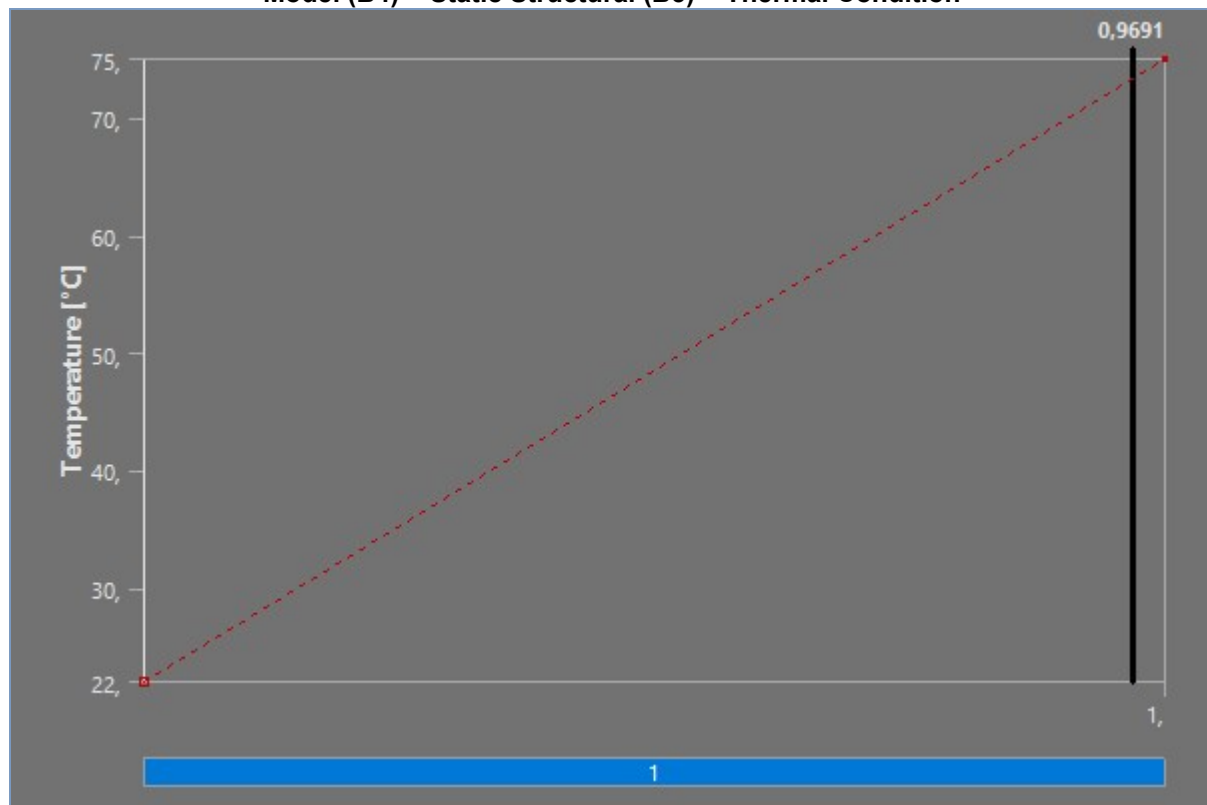


FIGURE 3
Model (B4) > Static Structural (B5) > Pressure

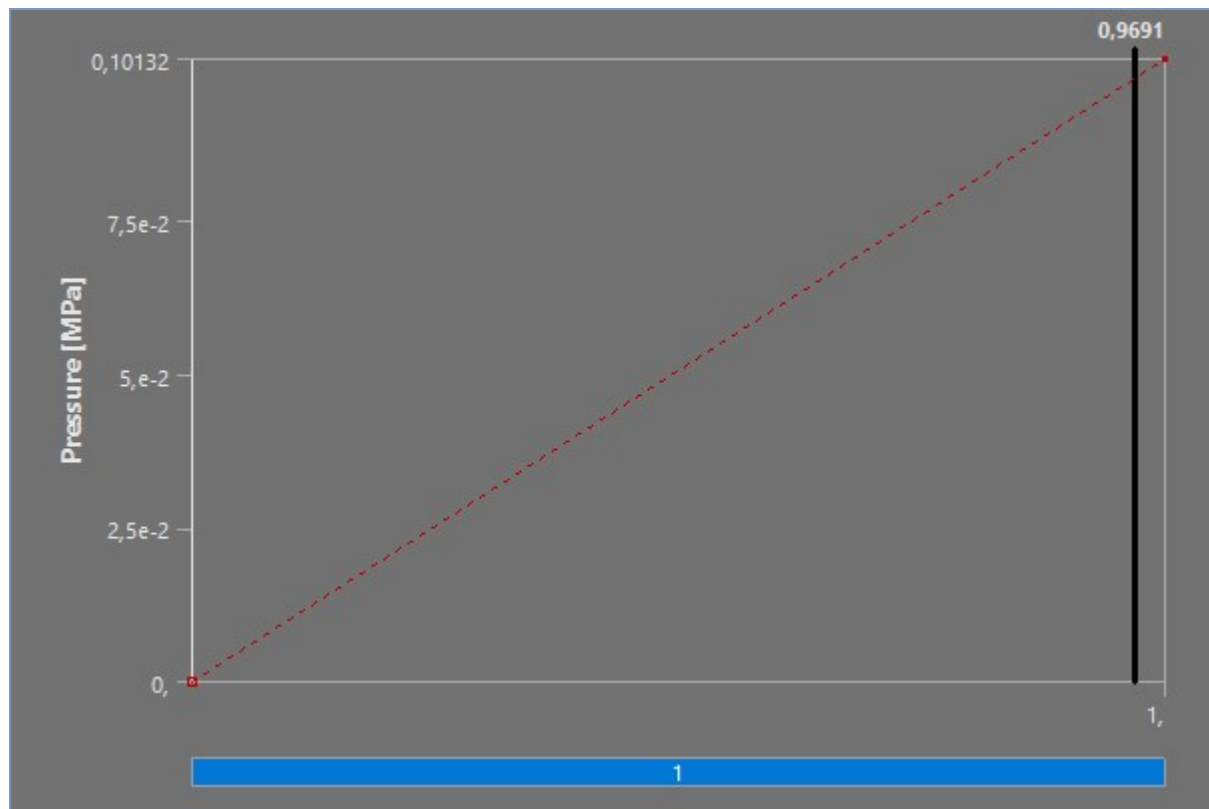
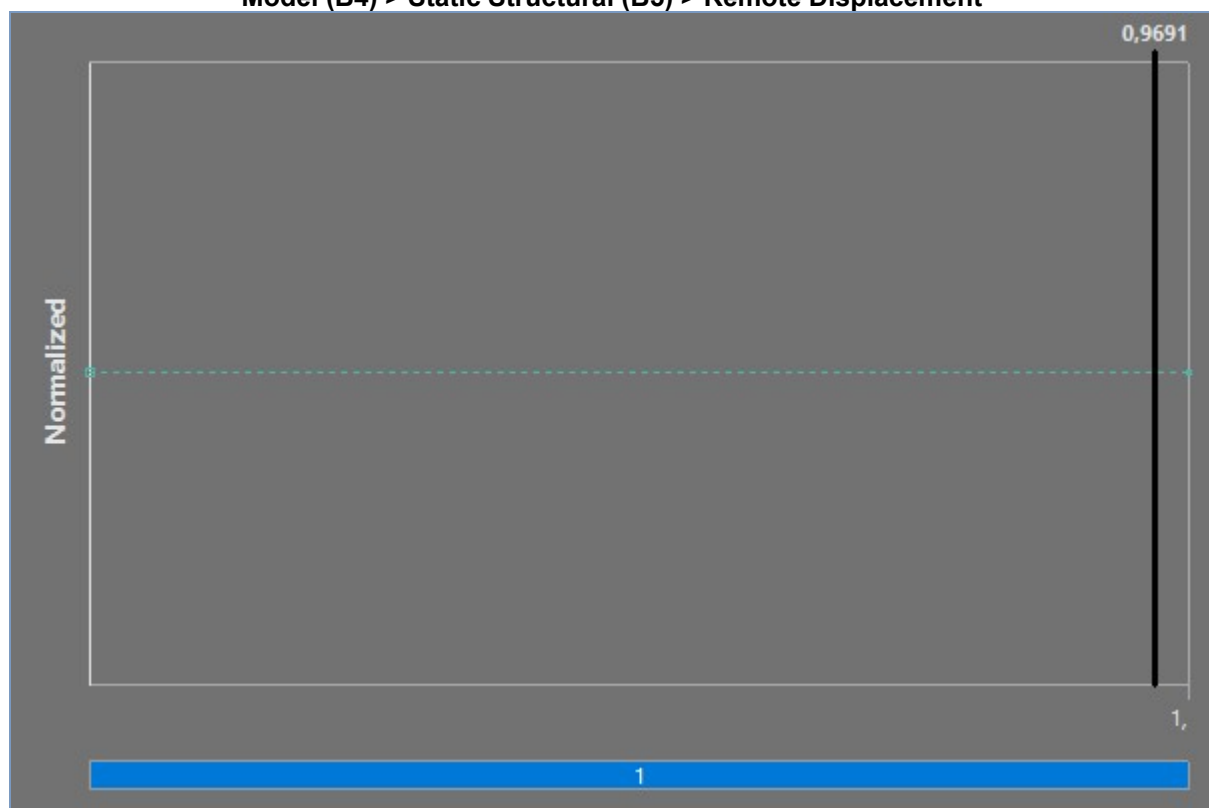


FIGURE 4
Model (B4) > Static Structural (B5) > Remote Displacement



Solution (B6)

TABLE 15
Model (B4) > Static Structural (B5) > Solution

Object Name	<i>Solution (B6)</i>
State	Solved

Adaptive Mesh Refinement	
Max Refinement Loops	1,
Refinement Depth	2,
Information	
Status	Done
MAPDL Elapsed Time	7 m 42 s
MAPDL Memory Used	938, MB
MAPDL Result File Size	609,31 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 16
Model (B4) > Static Structural (B5) > Solution (B6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	5
Update Interval	2,5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 17
Model (B4) > Static Structural (B5) > Solution (B6) > Results

Model (24) - Static Structural (25) - Solution (26) - Results		
Object Name	Total Deformation	Equivalent Stress
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Total Deformation	Equivalent (von-Mises) Stress
By	Time	
Display Time	Last	
Separate Data by Entity	No	
Calculate Time History	Yes	
Identifier		
Suppressed	No	
Results		
Minimum	2,0278 mm	7,992e-003 MPa
Maximum	13,637 mm	17,075 MPa
Average	8,6018 mm	2,2724 MPa
Minimum Occurs On	max_pressure Вырез-Вытянуть2	
Maximum Occurs On	max_pressure Вырез-Вытянуть2	
Minimum Value Over Time		
Minimum	0,21132 mm	1,0781e-005 MPa
Maximum	2,0278 mm	7,992e-003 MPa
Maximum Value Over Time		

Minimum	1,4093 mm	1,0506 MPa
Maximum	13,637 mm	17,075 MPa
Information		
Time	1, s	
Load Step	1	
Substep	14	
Iteration Number	49	
Integration Point Results		
Display Option		Averaged
Average Across Bodies		No

FIGURE 5
Model (B4) > Static Structural (B5) > Solution (B6) > Total Deformation

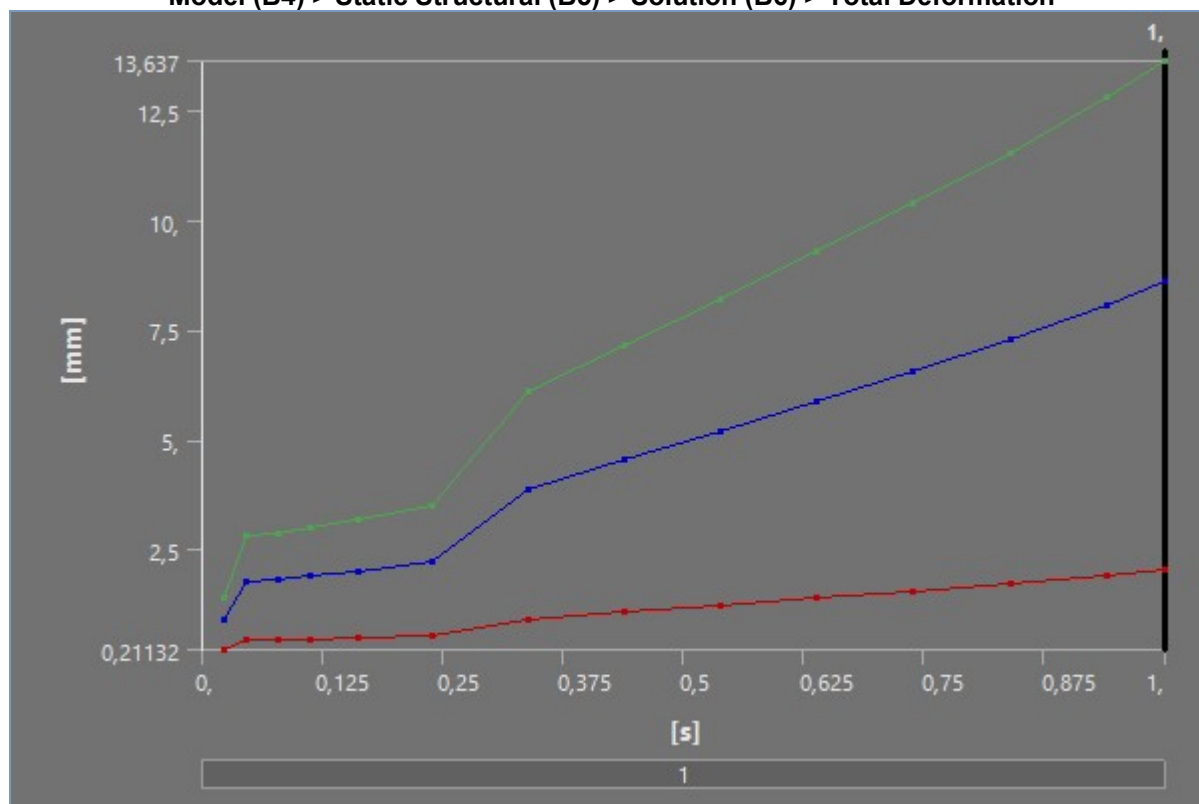


TABLE 18
Model (B4) > Static Structural (B5) > Solution (B6) > Total Deformation

Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
2,25e-002	0,21132	1,4093	0,89358
4,5e-002	0,42028	2,8035	1,7774
7,875e-002	0,43151	2,879	1,8253
0,1125	0,44629	2,9786	1,8884
0,16313	0,47494	3,1714	2,0106
0,23906	0,52684	3,5207	2,2318
0,33906	0,91053	6,0886	3,8571
0,43906	1,0686	7,152	4,5287
0,53906	1,2266	8,2166	5,2003
0,63906	1,3855	9,2893	5,8758
0,73906	1,548	10,388	6,5661
0,83906	1,7195	11,549	7,2943
0,93906	1,9041	12,799	8,0772
1,	2,0278	13,637	8,6018

FIGURE 6

Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress

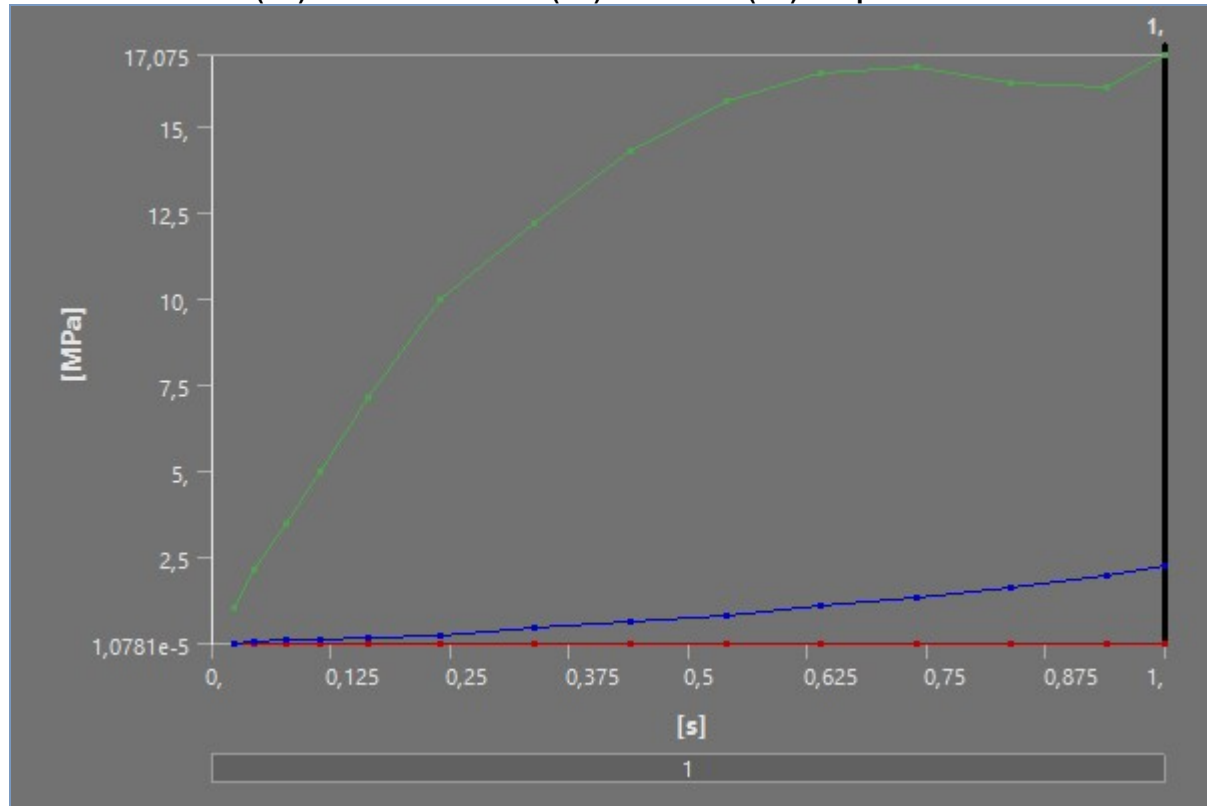


TABLE 19

Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
2,25e-002	1,0781e-005	1,0506	2,3002e-002
4,5e-002	3,7368e-004	2,1528	8,6098e-002
7,875e-002	7,3768e-005	3,4848	9,0776e-002
0,1125	3,7567e-004	4,9861	0,11356
0,16313	1,0407e-003	7,1301	0,16413
0,23906	2,0245e-003	10,009	0,25504
0,33906	2,2252e-003	12,187	0,44718
0,43906	2,8073e-003	14,31	0,62902
0,53906	3,1359e-003	15,76	0,83893
0,63906	2,991e-003	16,556	1,0773
0,73906	2,8524e-003	16,715	1,3436
0,83906	3,5486e-003	16,258	1,6381
0,93906	4,9993e-003	16,132	1,9908
1,	7,992e-003	17,075	2,2724

TABLE 20

Model (B4) > Static Structural (B5) > Solution (B6) > Stress Safety Tools

Object Name	<i>Stress Tool</i>
State	Solved
Definition	
Theory	Max Equivalent Stress
Stress Limit Type	Tensile Yield Per Material

TABLE 21

Model (B4) > Static Structural (B5) > Solution (B6) > Stress Tool > Results

Object Name	<i>Safety Factor</i>
State	Solved
Scope	

Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Safety Factor
By	Time
Display Time	Last
Separate Data by Entity	No
Calculate Time History	Yes
Identifier	
Suppressed	No
Integration Point Results	
Display Option	Averaged
Average Across Bodies	No
Results	
Minimum	4,5331
Minimum Occurs On	max_pressure Вырез-Вытянуть2
Minimum Value Over Time	
Minimum	4,5331
Maximum	15,
Maximum Value Over Time	
Minimum	15,
Maximum	15,
Information	
Time	1, s
Load Step	1
Substep	14
Iteration Number	49

FIGURE 7
Model (B4) > Static Structural (B5) > Solution (B6) > Stress Tool > Safety Factor

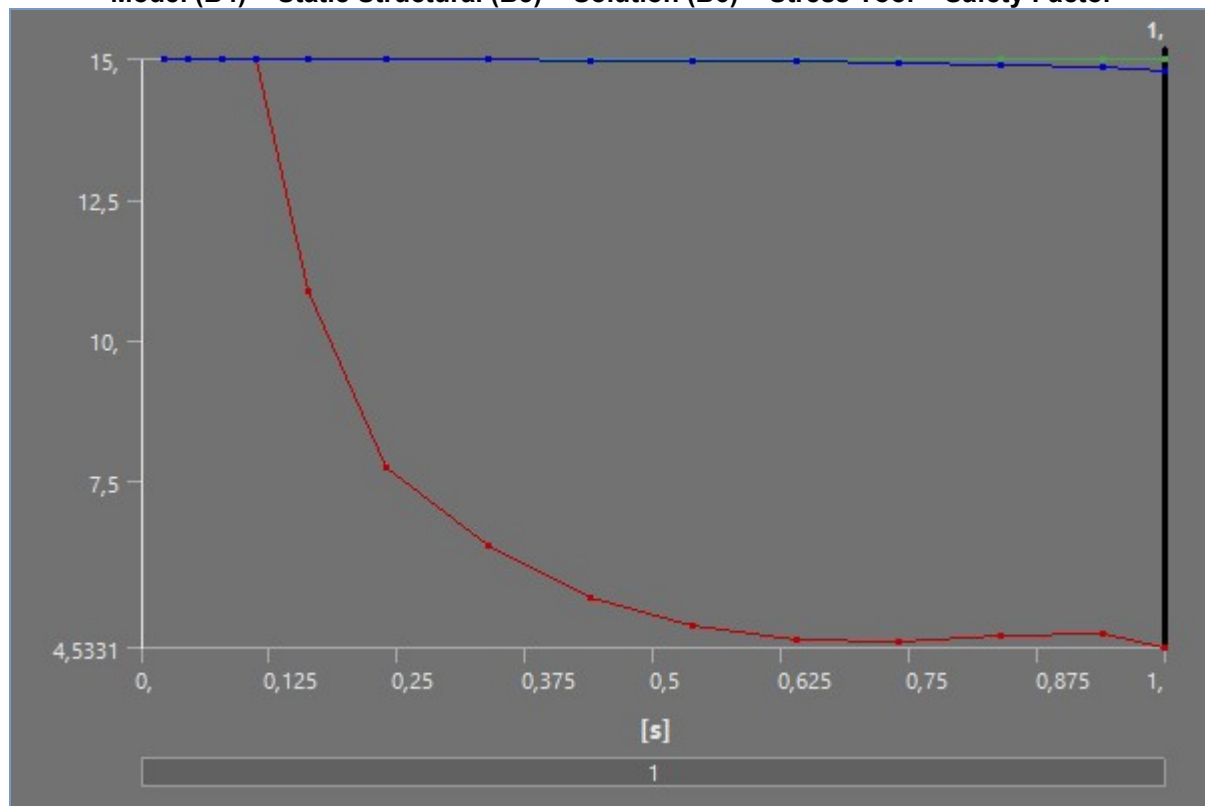


TABLE 22
Model (B4) > Static Structural (B5) > Solution (B6) > Stress Tool > Safety Factor

Time [s]	Minimum	Maximum	Average
2,25e-002	15,	15,	15,
4,5e-002			
7,875e-002			
0,1125			
0,16313	10,855		14,997
0,23906	7,7333		
0,33906	6,3511		
0,43906	5,409		
0,53906	4,9113		
0,63906	4,675		
0,73906	4,6306		
0,83906	4,7607		
0,93906	4,798		
1,	4,5331		14,792

Material Data

PA12-CF10

TABLE 23
PA12-CF10 > Constants

Density	1,04e-006 kg mm ⁻³
Coefficient of Thermal Expansion	7,e-005 K ⁻¹

TABLE 24
PA12-CF10 > Color

Red	Green	Blue
235,	222,	222,

TABLE 25
PA12-CF10 > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
3311,	0,38	4598,6	1199,6	295,15
1500,	0,4	2500,	535,71	343,15

TABLE 26
PA12-CF10 > Tensile Yield Strength

Tensile Yield Strength MPa
77,4

TABLE 27
PA12-CF10 > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
90,